

```

1 clear;
2 clc;
3 close all;
4
5 %% Total available bandwidth (MHz)
6 BW_total = 100;
7
8 %% Slice Requirements
9 % eMBB : High throughput
10 % URLLC : Ultra low latency, high priority
11 % mMTC : Large number of devices, small packets
12
13 req_eMBB = 50; % Required Mbps
14 req_URLLC = 20; % Required Mbps
15 req_mMTC = 10; % Required Mbps
16
17 % Priority:
18 % 1 = Low
19 % 2 = Medium
20 % 3 = High
21 priority = [2 3 1];
22
23 %% Dynamic Bandwidth Allocation
24
25 req = [req_eMBB req_URLLC req_mMTC];
26
27 % Calculate priority weights
28 priority_weights = priority ./ sum(priority);
29
30 % Allocate available bandwidth according to priority
31 BW_alloc = BW_total * priority_weights;
32
33 %% Spectral Efficiency
34 SE_eMBB = 6; % bits/s/Hz
35 SE_URLLC = 4; % bits/s/Hz
36 SE_mMTC = 1; % bits/s/Hz
37

```

Run

Save

Output

Input

==== Network Slicing Simulation Results ====

Allocated Bandwidth (MHz):

eMBB = 33.33 MHz

URLLC = 50.00 MHz

mMTC = 16.67 MHz

Throughput (Mbps):

eMBB = 200.00 Mbps

URLLC = 200.00 Mbps

mMTC = 16.67 Mbps

Latency (ms):

eMBB = 24.07 ms

URLLC = 1.30 ms

mMTC = 62.91 ms

[Execution complete with exit code 0]

Dynamic Bandwidth Allocation for 3 Network Slices

